Robust Target Presence Detection in Mixed Audio Using ConvTasNet

Elvin Rustamov Adalat Pashayev

April 3, 2026

Abstract

This article presents the design, implementation, and evaluation of a target presence detection system for mixed audio. The goal is not full speech recognition or source transcription, but binary decision making: given a mixed audio signal, determine whether a predefined target sound is present. A ConvTasNet-based neural model is used as the main backbone, followed by a lightweight classification head that maps the network output to a target-presence probability. A synthetic dataset was built from raw target and non-target waveform recordings, resampled to 8 kHz and segmented into 2-second clips. The system was trained and evaluated end to end on waveform inputs. On the held-out test set, the detector achieved an accuracy of 0.9555, precision of 0.9796, recall of 0.9673, specificity of 0.8919, and F1-score of 0.9735 at the default threshold of 0.50. Additional threshold optimization showed that the best F1-score of 0.9746 was obtained at a threshold of 0.58. A robustness experiment was also conducted by progressively amplifying the non-target component from 10 dB to 100 dB. The model remained effective under moderate interference but degraded substantially under extreme non-target amplification, with recall dropping from 0.74 at 10 dB to 0.06 at 100 dB. These results show that the proposed system is strong on the in-distribution test set and provide a clear baseline for future improvements in robustness and generalization.

1 Introduction

Target presence detection in audio is a practical and important problem in machine listening. In many real systems, the objective is not to fully transcribe speech or separate every sound source, but simply to answer a binary question: *is the target present in this audio mixture or not?* This kind of decision problem appears in surveillance, trigger-word systems, event monitoring, audio retrieval, source activity detection, and selective listening systems.

In this project, the target class can in principle be changed. However, the concrete experimental setup in this study uses piano audio as the target and several non-target audio sources as distractors. The task is therefore defined as follows:

> Given a mixed waveform, predict whether the target source is present.

This formulation is narrower and more appropriate than speech recognition. It is a binary detection problem built on raw audio waveforms. The system receives a waveform and outputs a single probability score, which is thresholded to produce a positive or negative prediction.

The work was built from scratch in a step-by-step manner. The full pipeline includes raw data organization, dataset generation, preprocessing, model construction, training, evaluation, threshold optimization, and robustness testing. A ConvTasNet-based architecture was chosen because it is well suited for raw waveform modeling and has a strong track record in source separation tasks. Instead of using ConvTasNet only for separation, this work adapts it into a detection architecture by attaching a classification head to the network output.

2 Problem Formulation

Let $x \in \mathbb{R}^T$ denote a mixed audio waveform of fixed length T. The target presence detection problem can be written as binary classification:

$$y = \begin{cases} 1, & \text{if the target source is present in the mixture} \\ 0, & \text{otherwise} \end{cases}$$

The model estimates a probability

$$\hat{p} = P(y = 1 \mid x)$$

and the final decision is obtained by thresholding:

$$\hat{y} = \begin{cases} 1, & \text{if } \hat{p} \geq \tau \\ 0, & \text{otherwise} \end{cases}$$

where τ is the decision threshold.

At the default operating point, $\tau = 0.50$ was used. A full threshold sweep was later performed to find the best threshold under different optimization criteria such as F1-score, precision, and recall.

3 Raw Data and Dataset Construction

3.1 Raw Data Organization

The raw audio data was organized into two categories:

- **Target**: one combined piano waveform file
- **Non-target**: combined waveform files containing background noise, male speech, and female speech

The corresponding raw folder structure was:

```
TargetPresenceDetection/
  data/
    raw/
      target/
        target_piano_combined.wav
      non_target/
```

```
noise_combined.wav
ted_female_combined.wav
ted_male_combined.wav
```

This organization made it straightforward to create a generic target-vs-non-target dataset generation process.

3.2 Preprocessing

All waveforms were standardized before dataset generation. The preprocessing steps were:

1. Load waveform from disk
2. Convert stereo audio to mono by channel averaging if needed
3. Resample all files to 8000 Hz
4. Apply peak normalization
5. Split the recordings into fixed-duration clips

The choice of 8000 Hz was deliberate. Higher sampling rates such as 16 kHz increase temporal resolution but also make raw waveform training heavier, slower, and more memory intensive. Since the task in this project is target presence detection rather than high-fidelity audio reconstruction, 8 kHz provides a practical trade-off between information retention and training efficiency.

Each audio clip was fixed to 2 seconds. Therefore, each model input contains:

$$T = 8000 \times 2 = 16000 \text{ samples}$$

3.3 Synthetic Sample Generation

The dataset was generated synthetically from the target and non-target clips. Three types of examples were created:

- **Target-only**: only the target clip is present, label $y = 1$
- **Non-target-only**: only the non-target clip is present, label $y = 0$
- **Mixed**: target and non-target are combined, label $y = 1$

The waveform mixing process is represented by:

$$x_{mix} = \alpha x_{target} + \beta x_{non-target}$$

where α and β are random gain factors sampled during dataset generation. In the implementation, the target gain was sampled from $[0.7, 1.0]$ and the non-target gain from $[0.3, 0.9]$ for normal training mixtures.

This design makes the detector learn both:

1. positive target evidence under interference, and
2. negative samples where the target is absent.

3.4 Train, Validation, and Test Splits

The dataset was split into train, validation, and test subsets. The final sample counts were:

- **Training**: 3632 samples
- **Validation**: 472 samples
- **Test**: 472 samples

From the test confusion matrix, the test set contains 398 positive examples and 74 negative examples. This shows that the test split is imbalanced toward positive samples, which is important when interpreting some metrics, especially precision-recall behavior.

4 Model Architecture

4.1 Why ConvTasNet

ConvTasNet was selected as the backbone because it operates directly on raw waveforms and is known to learn strong temporal representations. It was originally designed for source separation, but that property is useful here as well: the model can learn an internal representation that emphasizes the target-like component of the mixture before a classification decision is made.

This is more expressive than using a shallow classifier directly on raw waveforms and also more flexible than a simple energy threshold on the input waveform.

4.2 Architecture Used in This Work

The architecture consists of two conceptual parts:

1. **ConvTasNet separator backbone**: accepts a waveform mixture and produces an estimated target-like waveform
2. **Classification head**: converts the ConvTasNet output into a binary target-presence score

In practice, the forward pass can be summarized as:

$$x \to \text{ConvTasNet}(x) \to \hat{s} \to \text{Pooling} + \text{Linear Layers} \to z \to \sigma(z)$$

where:

- x is the input mixture,
- $\hat{s}$ is the estimated target-like waveform,
- z is the classification logit,
- $\sigma(\cdot)$ is the sigmoid function.

The classifier head uses adaptive average pooling followed by fully connected layers. This makes the system compact while still allowing the ConvTasNet backbone to learn rich raw-audio features.

5 Training Procedure

5.1 Optimization

The detector was trained using binary cross-entropy with logits:

$$\mathcal{L} = -\left[y \log(\sigma(z)) + (1-y)\log(1-\sigma(z))\right]$$

This is the standard loss for binary classification when the network outputs logits.

The optimizer was Adam with learning rate:

$$\eta = 10^{-3}$$

The batch size was increased to 20 after confirming that the hardware could handle larger batches stably. The total number of training epochs was 10.

5.2 Training Logs

The observed training and validation results were:

Epoch	Train Loss	Validation Loss	Validation Accuracy
1	0.3994	0.2948	0.8602
2	0.2694	0.2106	0.9174
3	0.2178	0.1620	0.9428
4	0.1895	0.1636	0.9407
5	0.1672	0.1738	0.9343
6	0.1522	0.1658	0.9364
7	0.1344	0.1605	0.9301
8	0.1304	0.1544	0.9364
9	0.1172	0.2007	0.9153
10	0.1003	0.1749	0.9322

Table 1: Training dynamics for the ConvTasNet-based detector.

These results show a healthy optimization process. Training loss decreases steadily, and validation loss also decreases overall, although mild overfitting begins to appear after the middle epochs. The best validation loss was observed at epoch 8.

6 Evaluation Methodology

6.1 Primary Metrics

The following standard binary classification metrics were used:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

6.2 Test Set Results at the Default Threshold

Using the default threshold $\tau = 0.50$, the held-out test set produced the following confusion counts:

- True Positives (TP): 385
- True Negatives (TN): 66
- False Positives (FP): 8
- False Negatives (FN): 13

The corresponding performance metrics are shown in Table 2.

Metric	Value
Accuracy	0.9555
Precision	0.9796
Recall	0.9673
Specificity	0.8919
F1-score	0.9735

Table 2: Held-out test performance at threshold 0.50.

These values indicate that the detector is already very strong as a baseline system. Precision is extremely high, which means that positive predictions are usually correct. Recall is also high, which means the system misses relatively few true target-containing samples.

6.3 ROC and Precision–Recall Analysis

Receiver Operating Characteristic (ROC) and Precision–Recall (PR) curves were generated from the saved probability predictions.

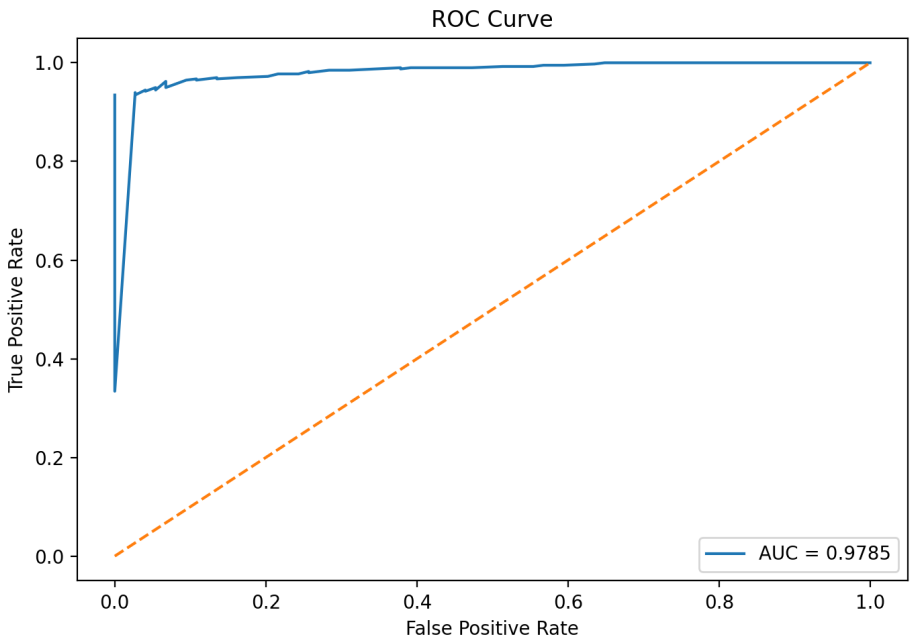

Figure 1: ROC curve for the ConvTasNet-based detector. The computed ROC AUC is 0.9785, indicating excellent separability between target-present and target-absent examples.

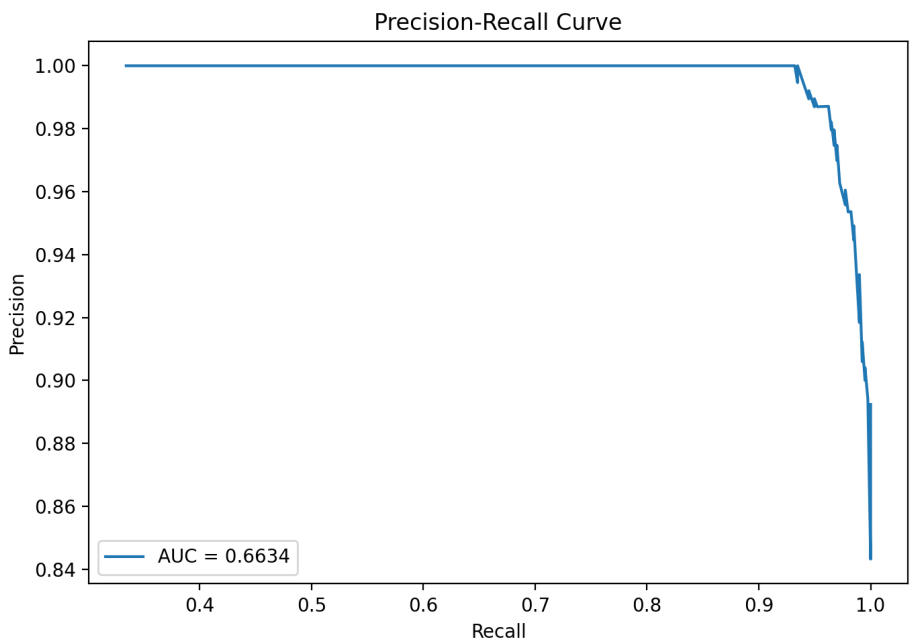

Figure 2: Precision–Recall curve for the ConvTasNet-based detector. The curve remains in a high-precision region across a broad recall range, which is consistent with the strong test-set precision observed at the default threshold.

The ROC AUC of 0.9785 confirms that the model separates the two classes very well over a wide range of thresholds.

The PR curve stays near very high precision across high recall values, which qualitatively indicates strong positive-class performance. The numerical PR area reported by the custom

script should be interpreted with care, because PR AUC is sensitive to interpolation method and class imbalance. In this dataset, the positive class is dominant on the test set, and the curve shape itself is more informative than the raw scalar alone.

6.4 Threshold Optimization

A threshold scan over 0.01 to 0.99 was performed using the saved test probabilities. The optimal threshold according to F1-score was:

- Best threshold by F1: $\tau = 0.58$
- TP = 383
- TN = 69
- FP = 5
- FN = 15
- Accuracy = 0.9576
- Precision = 0.9871
- Recall = 0.9623
- Specificity = 0.9324
- F1-score = 0.9746

This threshold slightly outperforms the default threshold of 0.50. It improves both specificity and F1-score while maintaining very strong recall.

Two additional operating points are useful:

- **Best precision** at $\tau = 0.87$, yielding precision = 1.0000
- **Best recall** at $\tau = 0.01$, yielding recall = 1.0000

This confirms that the detector can be tuned depending on the application. If false positives are costly, a higher threshold can be used. If missing the target is unacceptable, a lower threshold can be selected.

7 Robustness Testing

7.1 Motivation

A strong in-distribution test score does not necessarily imply robustness under severe interference. To evaluate the detector more thoroughly, a dedicated stress test was performed.

The idea of the stress test was simple: create additional synthetic mixtures and progressively amplify the non-target component by large positive dB values, then measure how detection performance changes.

7.2 Stress Test Design

For each gain level, 100 samples were generated:

- 50 samples with the target present
- 50 samples with only non-target audio

  The non-target component was amplified by:

$$10, 20, 30, \ldots, 100 \text{ dB}$$

A crucial detail in this experiment is that **the final mixture was not renormalized after applying the gain**. This preserves the actual effect of increasing non-target amplitude. If renormalization were applied after mixing, the gain experiment would be much less meaningful.

7.3 Stress Test Results

The results are shown in Table 3.

Gain (dB)	Accuracy	Precision	Recall	F1-score
10	0.74	0.7400	0.7400	0.7400
20	0.66	0.8077	0.4200	0.5526
30	0.56	0.6875	0.2200	0.3333
40	0.51	0.5714	0.0800	0.1404
50	0.48	0.3333	0.0400	0.0714
60	0.51	0.5556	0.1000	0.1695
70	0.51	0.6000	0.0600	0.1091
80	0.50	0.5000	0.0600	0.1071
90	0.47	0.3333	0.0600	0.1017
100	0.48	0.3750	0.0600	0.1034

Table 3: Stress-test performance as the non-target gain increases.

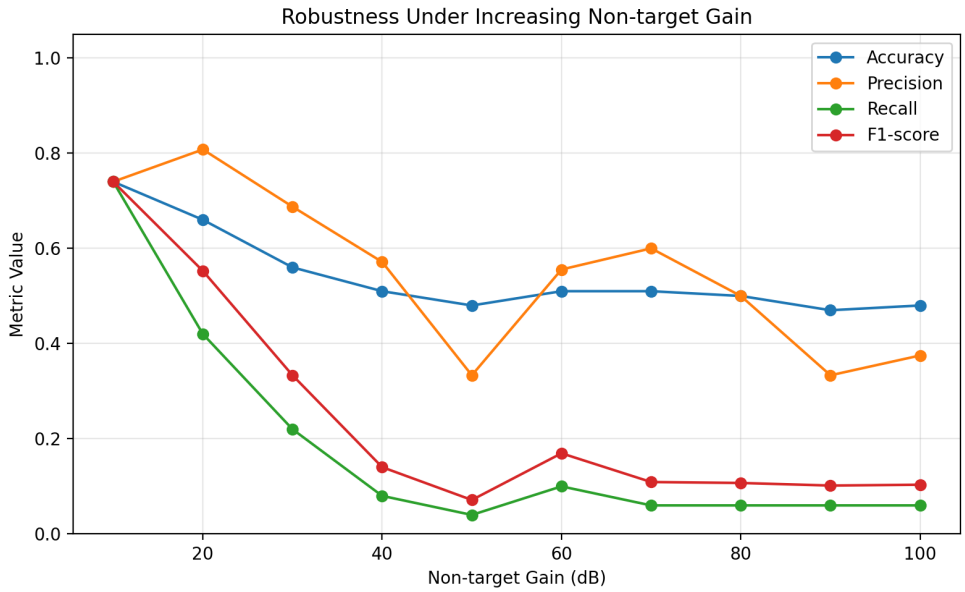

Figure 3: Robustness plot showing the degradation of accuracy, precision, recall, and F1-score as the non-target signal is amplified from 10 dB to 100 dB.

7.4 Interpretation of the Robustness Results

The robustness experiment reveals a very important property of the system.

At moderate interference (10 dB), the model is still usable, with balanced precision and recall. However, performance drops quickly as the non-target gain increases. The clearest failure mode is the collapse in recall:

- Recall = 0.74 at 10 dB
- Recall = 0.42 at 20 dB
- Recall = 0.22 at 30 dB
- Recall = 0.08 at 40 dB
- Recall = 0.06 at 100 dB

This means that under extreme interference, the detector increasingly fails to detect the target even when the target is actually present. In other words, the system becomes dominated by false negatives. This is exactly what one would expect when the non-target source overwhelms the target waveform.

The robustness test is therefore useful for two reasons:

1. It shows that the baseline model is strong under the original test distribution.
2. It shows that extreme amplitude mismatch remains a major weakness.

8 Discussion

8.1 What Worked Well

Several aspects of the system worked well:

- The end-to-end pipeline from raw audio to evaluation was successfully built and executed.
- Raw-waveform learning with ConvTasNet was effective for this binary task.
- The test-set F1-score above 0.97 demonstrates strong in-distribution performance.
- Threshold optimization further improved the detector beyond the default setting.
- The evaluation setup was extended beyond accuracy to include ROC, PR, and robustness testing.

8.2 Observed Limitations

Despite the strong results, the system has several limitations:

- The target data comes from a single combined target source file, which limits diversity.
- The non-target data is also relatively small and comes from only a few source types.
- The dataset is synthetic, which helps with controlled experimentation but may not fully represent real-world mixtures.
- The test set is imbalanced toward positive samples.
- The robustness results show that the detector struggles when the non-target amplitude becomes extreme.

8.3 Why the Robustness Gap Appears

The model was trained mainly on mixtures whose relative gains were sampled within moderate ranges. During stress testing, the non-target source was amplified far beyond the training regime. Since the training distribution did not frequently expose the network to these extreme ratios, the model naturally loses sensitivity to the target in such cases.

This suggests a clear path forward: robustness will likely improve if the training pipeline includes wider gain ranges, harder negative samples, and more varied interference conditions.

9 Future Work

Several improvements are natural extensions of this work:

1. **More diverse target and non-target recordings.** More speakers, more instruments, more background conditions, and more recording environments would improve generalization.
2. **Training with harder mixtures.** Including stronger non-target gains during training would make the model less surprised by extreme interference.

3. **Multi-task learning.** A combined objective using both classification loss and separation loss could encourage the ConvTasNet backbone to preserve more target-specific information.

4. **Class balancing.** Since the test set is strongly positive-heavy, balancing the dataset could provide a more stable operating point.

5. **Calibration and threshold selection.** The detector can be calibrated to different operating points depending on whether recall or precision is more important.

6. **Real-time deployment.** Since the model uses 8 kHz waveforms and a compact head, it may be a good candidate for practical streaming or edge deployment after optimization.

10 Conclusion

This work presented a complete target presence detection system for mixed audio based on a ConvTasNet backbone and a binary classification head. The project was built from raw waveform files, converted into a synthetic training dataset, and trained end to end on 8 kHz waveform segments. The final detector achieved excellent test performance, including 0.9555 accuracy and 0.9735 F1-score at the default threshold, with a further improvement to 0.9746 F1-score after threshold optimization.

Beyond the standard test set, the system was also evaluated under a dedicated robustness stress test in which the non-target signal was aggressively amplified. This additional analysis showed that although the detector is strong under the original evaluation distribution, it still degrades sharply under extreme interference. As a result, the project provides both a successful baseline and a clear roadmap for future work.

Overall, the study demonstrates that ConvTasNet can be repurposed successfully for target presence detection, not just source separation, and that raw-waveform neural models can produce highly accurate binary decisions in controlled mixed-audio scenarios.